

DHESIKAN DR

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PROFESSIONAL SUMMARY

Results-driven Aerospace Engineering student and researcher with core competency in Fluid Dynamics, Orbital Mechanics, and UAV Architecture. Skilled at bridging traditional aerospace engineering with computational technologies, including Robotics, UI/UX Design, and Full-Stack Development. Experience spanning space-debris mitigation using electrodynamic tethers, atmospheric observation UAVs, cold-gas propulsion systems, composite structures, and kinematic analysis. Interested in multidisciplinary aerospace and defense challenges focused on performance and sustainability.

EDUCATION

Hindustan Institute of Technology and Science (HITS) — Chennai, India

Bachelor of Technology in Aerospace Engineering | Expected 2027

Relevant Coursework: Engineering Mechanics, Thermodynamics, Fluid Dynamics, Strength of Materials, Engineering Graphics, Structural Analysis, Propulsion Systems.

TECHNICAL EXPERIENCE

Composite Structures & Manufacturing Intern — BRAHMASTRA AEROSPACE | 2025

- Completed an intensive technical internship focused on advanced manufacturing, maintenance, and structural validation of aerospace composite systems.
- Developed proficiency in fabrication of carbon-fiber and fiberglass airframe components using vacuum bagging and precision hand lay-up techniques.
- Optimized structural repair procedures for sandwich panels and monolithic laminates while maintaining aerospace weight constraints.
- Implemented Non-Destructive Testing (NDT) and visual inspection protocols to validate bond quality and laminate consolidation.

Robotics Intern — Wissensquelle | 2025

- Performed forward and inverse kinematic simulations for ARISTO, 6-DOF industrial, and STANFORD robots using RoboAnalyzer.
- Conducted kinematic analysis of Gantry Robots using RoboAnalyzer and CPRog to improve pick-and-place precision.
- Synthesized and fabricated hybrid advanced composites and tested material strength and structural integrity for aerospace applications.

Full-Stack Developer — Project: AxonShare

- Engineered offline file-sharing software using Python and web technologies for secure, high-speed data transfer without internet connectivity.
- Implemented user-centric UI/UX design for seamless navigation and cross-platform compatibility in air-gapped systems.

AEROSPACE ENGINEERING PROJECTS

ZENITH-950: High-Stability Research UAV | 2026

- Designed and developed a high-wing research UAV platform optimized for stable atmospheric observation and high-fidelity data collection.
- Optimized high-wing geometry and tail configuration to maximize lateral stability and improve lift characteristics in low-Reynolds-number flight regimes.
- Performed weight and balance analysis for optimal Center of Gravity (CG) placement and steady-state flight.

CGT IRIS DELTA V: Cold Gas Propulsion System | 2025

- Designed a regulated cold-gas propulsion system for 6U CubeSat applications, targeting precise orbital station-keeping and maneuvering.
- Optimized nozzle geometry using CFD to achieve a target thrust of 50 mN with nitrogen as the propellant.
- Conducted Delta-V budget analysis and structural pressure-vessel simulations for orbital mission survival.

RESEARCH & SCIENTIFIC ACHIEVEMENTS

Silver Honor — International Astronomy and Astrophysics Competition (IAAC) | 2025

- Received Silver Honor in a high-level international competition and advanced through the Qualification and Pre-Final rounds.
- Solved problems involving orbital mechanics, stellar evolution, and galactic physics, advancing to the Global Final stage.

Eradication of Space Debris using Electrodynamic Tethers | 2025

- Investigated Lorentz-force-based orbital decay mechanisms as a sustainable approach for removing non-functional satellites from Low-Earth Orbit (LEO).
- Analyzed tether-based propulsion feasibility for reducing risks associated with Kessler Syndrome in high-traffic orbital planes.

Hybrid Rocket Engine Development for Space Launch | 2025

- Researched propulsion efficiency using solid fuels with liquid oxidizers to improve safety, restart capability, and throttle control.
- Optimized propellant grain geometry for stable and predictable thrust profiles in small-scale satellite launch applications.

TECHNICAL SKILLS

- Engineering Tools: AutoCAD, RoboAnalyzer, CPROg, Composite Material Fabrication, CFD Analysis.
- Programming: Python, HTML, JavaScript, Full-Stack Web Development, Matplotlib.
- Design: UI/UX Design (Wireframing & Prototyping), Figma, Engineering Graphics.
- Soft Skills: Entrepreneurial Mindset, Strategic Planning, Technical Research, Agile Methodologies.

PROFESSIONAL INTERESTS

- Space Exploration: Orbital sustainability, cold-gas propulsion, and interplanetary architectures.
- UAV Dynamics: High-stability configurations and autonomous sensor integration for research.
- Astrophysics: Theoretical modeling of celestial phenomena and deep-space instrumentation.